make it

use it

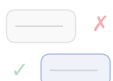
run and adapt it

Pretrain

next token on ~15T tokens

Scale

loss falls as a power law

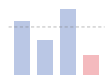


Post-train

base model to assistant

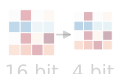
Prompt

steer it, no training



Evaluate

benchmarks and judges



Quantize

8B model: 16 → 5 GB

LoRA

train < 1% of weights